Pre-training Buys Ordering, Not Calibration: An EveNet Data-Efficiency Test for κ_λ Density-Ratio Learning in $HH \to b\bar{b}\tau^+\tau^-$ (CMS Full Simulation)

Working note — $b\bar{b}\tau^+\tau^-$ κ_λ NSBI project

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Gates G0 / G0.5 of `EVENET_FEASIBILITY_NOTE`; 90-run sweep, branch `fm`

Abstract

We test whether the pre-trained event-level foundation model (FM) EveNet [1] improves the data efficiency of *density-ratio* learning for a Higgs self-coupling (κ_λ) measurement via neural simulation-based inference (NSBI), using CMS full-simulation $HH \to b\bar{b}\tau^+\tau^-$ events. Three arms — fine-tuned, frozen-backbone, and from-scratch — identical in data, architecture and head and differing *only* in initialisation, were trained at six training fractions × five seeds (90 runs) and evaluated on one fixed held-out split against two axes fixed before unblinding: discrimination (weighted area under the receiver-operating characteristic curve, AUC) and ratio calibration (integral and shape closure). The result is a clean dissociation. On discrimination, pre-training wins decisively and consistently: +0.014–0.017 AUC at every fraction $\geq 1\%$ (many standard deviations), equivalent to a $\approx$**3–4.5**× data multiplier, with both pre-trained arms exceeding a feature-level gradient-boosted decision tree (BDT) ceiling. On calibration, pre-training buys *nothing*: no arm passes the closure gates at any fraction, the pre-registered adoption metric (closure left-shift) is undefined, and the arms trade places within seed errors. Seed-ensembling does not change this, because the closure offsets are same-sign systematic ($\mathbb{E}_{\text{ref}}[\hat{r}] - 1 \approx +0.13$–$0.29$) rather than zero-centred. The frozen arm matches fine-tuning everywhere, locating the value in the pre-trained *representation* rather than in backbone adaptation. We argue the calibration failure is a property of EveNet's noise-conditioned classification objective rather than of the task, note that the feature BDT reaches $|\text{IC}| = 0.018$ where EveNet reaches 0.13–0.17 *despite worse AUC*, and propose a single decisive ablation.

Contents

1 What was tested

Physics task. Learn per-event density ratios $r(x; \kappa_\lambda, \kappa'_\lambda)$ between κ_λ hypotheses for use in an unbinned NSBI likelihood. Discrimination between κ_λ points is driven by the m_{HH} lineshape and triangle–box interference, i.e. by a *continuous* deformation of one process's spectrum rather than by separation of two distinct processes.

Data. CMS 2024 full-simulation $HH \to b\bar{b}\tau^+\tau^-$ Powheg samples, passed through the KIT CROWN analysis chain (published $\mu\tau_h/e\tau_h$ selection), converted to EveNet's stock 7-feature token schema: four tokens per event (b_1, b_2, τ_h, ℓ), missing transverse energy in the global conditioning block, weights = `genWeight` × `puweight`. Requiring a valid $H \to b\bar{b}$ candidate drops 18.6–24.5% of events (κ_λ-dependent: softer $\kappa_\lambda = 5$ spectrum ⇒ more failed second-b reconstruction). Class 0 = $\kappa_\lambda = 1$ (SM reference, 488k events), class 1 = $\kappa_\lambda = 5$ (339k). The merged, shuffled sample is split 80/10/10 into 661.9k train / 82.7k validation / 82.7k test; the training fractions below multiply the *train* split, and **all** runs share the one fixed 82.7k-event test split.

The contrast. Three arms differing only in initialisation:

finetune
20M-parameter pre-trained checkpoint, all weights trainable.

frozen
same checkpoint, backbone (GlobalEmbedding, PET, ObjectEncoder) frozen, heads trainable — isolates the *representation* from optimiser effects.

scratch
random initialisation — the no-FM control.

Each at training fractions $\{0.003, 0.01, 0.02, 0.05, 0.1, 1.0\}$ (the fractions apply to the 661.9k-event training split, i.e. ≈ 2.0k to 662k training events) × 5 seeds = 90 runs. Options files are the authors' own downstream recipes (fine-tune and from-scratch variants of the Exotic-Higgs study), vendored unchanged apart from the freeze flags.

External bar. A gradient-boosted decision tree on the 10 high-level features behind the published unbinned result, measured at full statistics on the same physics (Sec. 6).

2 Pre-registration record

The following were fixed *before* any sweep output existed, and are reproduced here unchanged:

(P1) **Decision primacy (2026-07-12).** The adoption metric is the *closure left-shift*: the smallest training fraction at which an arm's seed-mean passes both gates, compared between finetune and scratch; their ratio is the equivalent-data multiplier. A left-shift is an adoption-grade win *even at AUC parity*; an AUC-only win with failing closure is *not* a win. AUC serves as a secondary no-regression guard.

(P2) **Gates.** $|\mathrm{IC}| \leq \max(1\%, 3 \times \mathrm{stat})$ *and* stat-debiased shape RMS ≤ 0.05 (= the analysis `CLOSURE_TOL`), with $\chi^2/\mathrm{ndf} \approx 1$ indicating noise-only residuals.

(P3) **Checkpoint selection (2026-07-14).** Every arm is evaluated at its best-validation-loss checkpoint (EveNet's predictor otherwise loads the final, i.e. most overfit, epoch).

(P4) **Fallback.** Should no arm pass the gates anywhere — leaving the left-shift undefined — the secondary reading is relative closure at matched fractions, reported as such.

The fallback (P4) is the branch that actually materialised.

3 Metrics

3.1 The closure identity

For scores p, labels y and weights w on the test split, the ratio estimate is $\hat{r} = \frac{p}{1-p} \cdot f_{\mathrm{prior}}$, with f_{prior} the inverse of the class prior the training loss effectively used (measured, Sec. 8). Everything below rests on one identity. Average the *true* ratio over events drawn from the reference:

$$\mathbb{E}_{\mathrm{ref}}[r] \;=\; \int p_{\mathrm{ref}}(x)\,\frac{p_{\mathrm{hyp}}(x)}{p_{\mathrm{ref}}(x)}\,\mathrm{d}x \;=\; \int p_{\mathrm{hyp}}(x)\,\mathrm{d}x \;=\; 1. \tag{1}$$

The reference density cancels against the denominator and what remains is a normalised probability density. So the true ratio, averaged over reference events, is *exactly* one — for any pair of hypotheses, with no free parameters. Intuitively, $r(x)$ says "this event is r times more likely under the hypothesis"; over the reference ensemble the regions where $r > 1$ and $r < 1$ must balance precisely. On a weighted Monte Carlo sample the expectation becomes the self-normalised weighted average $\sum_{i\in\mathrm{ref}} w_i \hat{r}_i / \sum_{i\in\mathrm{ref}} w_i$, which converges to Eq. (1) at rate $1/\sqrt{n_{\mathrm{eff}}}$ — hence every closure number carries a statistical error and is "consistent with 1", never "equal to 1".

3.2 The two metrics

Integral closure, $\mathrm{IC} = \mathbb{E}_{\mathrm{ref}}[\hat{r}] - 1$

Definition: the weighted mean of $\hat{r}$ over held-out reference events, minus one; quoted with its weighted standard error. *Intuition:* reweight the reference sample by $\hat{r}$ and ask whether you recover the *total* hypothesis yield. $\mathrm{IC} = +0.20$ means the reweighted reference over-produces the hypothesis by 20%. It is the *normalisation* or "exchange rate" error of the ratio. Two-bin illustration: reference $(0.5, 0.5)$, hypothesis $(0.8, 0.2)$ gives true $r = (1.6, 0.4)$ and $\mathbb{E}_{\mathrm{ref}}[r] = 0.5(1.6) + 0.5(0.4) = 1$ exactly; a miscalibrated $\hat{r} = (2.0, 0.4)$ gives 1.2, i.e. $\mathrm{IC} = +20\%$.

Shape closure, $\mathrm{SC}_{\mathrm{rms}}$ **and** $\mathrm{SC}_{\mathrm{norm}}$

Definition: quantile-bin the score; in each bin compare the $\hat{r}$-reweighted reference mass A_b with the actual hypothesis mass B_b, forming $\mathrm{rel}_b = A_b/B_b - 1$; report the RMS over bins with each bin's statistical variance $(1/n_{\mathrm{eff},0} + 1/n_{\mathrm{eff},1})$ subtracted in quadrature, so finite-sample noise does not masquerade as bias. *Intuition:* IC can be satisfied by errors that cancel globally — too large here, too small there. The shape test therefore applies Eq. (1) *differentially*: after reweighting the reference by $\hat{r}$, it must become statistically indistinguishable from the hypothesis sample *bin by bin*, which is precisely what "being the density ratio" means. We bin in the classifier score because the score is monotone in $\hat{r}$ itself, so any miscalibration is forced to show up there.

Why the per-bin error is $1/n_{\mathrm{eff}}$**.** Both A_b and B_b are Monte Carlo sums over finite, *weighted* samples, so both fluctuate. For a weighted sum $S = \sum_i v_i$ over events whose presence is independent (Poisson), the variance is $\mathrm{Var}(S) = \sum_i v_i^2$, and therefore the *relative* variance is

$$\frac{\mathrm{Var}(S)}{S^2} = \frac{\sum_i v_i^2}{\big(\sum_i v_i\big)^2} \equiv \frac{1}{n_{\mathrm{eff}}}, \qquad n_{\mathrm{eff}} \equiv \frac{\big(\sum_i v_i\big)^2}{\sum_i v_i^2}, \tag{2}$$

which is precisely the definition of the effective sample size n_{eff} (Kish). For unweighted events ($v_i = 1$) it collapses to $n_{\mathrm{eff}} = N$ and to the familiar Poisson result $\sqrt{N}/N = 1/\sqrt{N}$; unequal weights always reduce it. Concretely, $v = (1,1,1,1)$ gives $n_{\mathrm{eff}} = 16/4 = 4$, whereas $v = (10,1,1,1)$ gives $169/103 = 1.6$: four events that behave statistically like 1.6, because a single event carries most of the sum. In our bins $v_i = w_i \hat{r}_i$ on the reference side — the reweighting inflates the weight spread, which is why bins at $\hat{r} \to \infty$ are dropped by the n_{eff} threshold rather than allowed to dominate — and $v_i = w_i$ on the hypothesis side. Since $\mathrm{rel}_b = A_b/B_b - 1$ is a ratio of estimates from two *independent* samples, the relative variances add: $\sigma_b^2 = 1/n_{\mathrm{eff},0} + 1/n_{\mathrm{eff},1}$.

Why the subtraction is in quadrature. Write $\mathrm{rel}_b = \beta_b + \varepsilon_b$, with β_b the genuine bias and ε_b zero-mean noise of variance σ_b^2. Then $\mathbb{E}[\mathrm{rel}_b^2] = \beta_b^2 + \sigma_b^2$ — the naive RMS is inflated by noise even for a perfectly calibrated model, and increasingly so at small statistics, exactly where the interesting comparisons live. Averaging over bins and subtracting the known σ_b^2 therefore estimates the bias RMS, $\mathrm{SC}_{\mathrm{rms}} = \sqrt{\max\big(\langle \mathrm{rel}^2 - \sigma^2 \rangle, 0\big)}$. The clipping at zero is why exact 0.000 entries appear in Table 3: they mean "no bias resolvable above the statistical floor", not literal perfection. Without the subtraction the metric rejects correct models — the first implementation did precisely that, failing an analytically calibrated toy in three seeds out of four. The same σ_b normalises the reported $\chi^2/\mathrm{ndf} = \langle (\mathrm{rel}_b/\sigma_b)^2 \rangle$, which is ≈ 1 when the residuals are pure noise and $\gg 1$ when they are not.

3.3 $\mathrm{SC}_{\mathrm{rms}}$ versus $\mathrm{SC}_{\mathrm{norm}}$

Mechanically. $\mathrm{SC}_{\mathrm{rms}}$ is computed from $\hat{r}$ as delivered; $\mathrm{SC}_{\mathrm{norm}}$ from $\hat{r}/\mathbb{E}_{\mathrm{ref}}[\hat{r}]$, i.e. after forcing the integral closure to zero. The difference is exactly the contamination of the shape metric by the normalisation error. Suppose $\hat{r} = c\, r_{\mathrm{true}}$ — a *pure* scale error with a perfect shape. Then in every bin $A_b/B_b = c$, so $\mathrm{rel}_b = c - 1$ for all b and $\mathrm{SC}_{\mathrm{rms}} \simeq |c-1| \simeq |\mathrm{IC}|$, even though nothing about the shape is wrong. Dividing by $\mathbb{E}_{\mathrm{ref}}[\hat{r}] = c$ first gives $\mathrm{SC}_{\mathrm{norm}} = 0$. The two raw numbers are therefore *not* independent, and $\mathrm{SC}_{\mathrm{rms}}$ alone cannot distinguish "the ratio is the wrong size" from "the ratio is the wrong function". Reading them together gives three clean signatures:

signature	$\lvert\mathrm{IC}\rvert$	$\mathrm{SC}_{\mathrm{rms}}$	$\mathrm{SC}_{\mathrm{norm}}$	diagnosis
calibrated	≈ 0	≈ 0	≈ 0	usable
pure scale error	$\lvert c-1\rvert$	$\approx \lvert\mathrm{IC}\rvert$	≈ 0	normalisation only — removable
shape distortion	any	large	large	deformed likelihood — not removable

Table 1: How to read the pair. Validated on synthetic data: a $\times 1.4$ scale error gives (0.399, 0.399, 0.000); a temperature-1.5 distortion gives (0.450, 0.694, 0.503).

Physically — why the distinction is not cosmetic. In an NSBI likelihood the contribution of process p enters as $\nu_p\, p_p(x)$, with ν_p the expected yield (from cross section $\times$ luminosity $\times$ acceptance, determined independently of the classifier) and $p_p(x) = r_p(x)\, p_{\mathrm{ref}}(x)$. If $\hat{r}_p = c\, r_p$, then

$$\nu_p\, \hat{p}_p(x) \;=\; \big(c\,\nu_p\big)\, \Big[\hat{p}_p(x)/c\Big], \tag{3}$$

i.e. the constant is algebraically indistinguishable from having mis-stated the *yield* by the factor c while leaving the normalised per-event shape untouched. Two consequences follow. First, it is *removable by construction*: rescaling $\hat{r}_p \to \hat{r}_p/\mathbb{E}_{\mathrm{ref}}[\hat{r}_p]$ on held-out reference events restores $\mathbb{E}_{\mathrm{ref}} = 1$ exactly, at the cost only of the statistical error on that single number. Second, uncorrected, it propagates into the fit *solely* through that process's normalisation, where it is degenerate with a rate nuisance parameter. For a coupling measurement this is emphatically not harmless — $\sigma(\kappa_\lambda)$ is exactly what carries the physics, so an uncorrected c biases κ_λ — but it is a one-parameter defect with a one-parameter cure, and it does not touch the discriminant.

A shape error is different in kind. There $\hat{r}(x)/r(x)$ varies with x, so events are mis-weighted *relative to one another*: the per-event likelihood is deformed, no single constant repairs it, and no rate parameter absorbs it, so the induced bias on κ_λ survives profiling. This is why we gate on both, and why $\mathrm{SC}_{\mathrm{norm}}$ — reported alongside, never instead of, the pre-registered pair — is the number that says whether a failure is repairable.

Which signature our data shows. At full statistics all three arms sit close to the "pure scale" row of Table 1: frozen $(\lvert\mathrm{IC}\rvert, \mathrm{SC}_{\mathrm{rms}}, \mathrm{SC}_{\mathrm{norm}}) = (0.171, 0.163, 0.044)$, finetune (0.133, 0.115, 0.074), scratch (0.156, 0.194, 0.084). The residual is predominantly a normalisation error with a comparatively small shape component — which is precisely the fingerprint of a mis-specified prior factor in the ratio conversion, and the reason the outstanding check of Sec. 7 targets exactly that.
$\mathrm{SC}_{\mathrm{norm}}$ is a *post-hoc* diagnostic added during analysis and is reported alongside, never instead of, the pre-registered gates. Validation: on synthetic data a pure $\times 1.4$ scale error gives $\mathrm{SC}_{\mathrm{rms}} = 0.399$ but $\mathrm{SC}_{\mathrm{norm}} = 0.000$, while a genuine temperature distortion gives 0.694 and 0.503 respectively.

What closure cannot catch. Both metrics test *self-consistency*, not *sufficiency*. A classifier trained on lossy features $\phi(x)$ converges to the exact ratio of the *marginalised* densities $p_{\mathrm{hyp}}(\phi)/p_{\mathrm{ref}}(\phi)$, which satisfies Eq. (1) identically and passes any test binned in a function of ϕ — the degenerate estimator $\hat{r} \equiv 1$ passes both metrics perfectly while carrying no information at all. Information loss does not appear as bias; it appears as a *flatter* ratio, i.e. as lost discrimination. Closure therefore catches lies, not omissions: it earns $\hat{r}$ the right to enter the likelihood, while the AUC axis (Sec. 4) is what shows the ratio is worth entering. This is also why a score-binned closure test cannot substitute for the analysis-level closure in physical observables, where the model is asked to reproduce distributions it was not free to choose the coordinates of.

4 Result 1 — discrimination (gate G0)

	single model (seed mean ± s.d., $n = 5$)			**5-seed ensemble**		
fraction	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.7427(57)	0.7498(80)	**0.7573(122)**	0.7576	0.7699	**0.7738**
0.01	0.7768(108)	**0.7793(76)**	0.7612(300)	**0.7945**	0.7935	0.7822
0.02	**0.7941(43)**	0.7931(80)	0.7804(19)	0.8050	**0.8056**	0.7877
0.05	0.8056(54)	**0.8099(22)**	0.7949(8)	0.8150	**0.8167**	0.8000
0.1	0.8163(22)	**0.8169(9)**	0.8003(25)	0.8214	**0.8217**	0.8075
1.0	**0.8313(2)**	**0.8313(5)**	0.8250(4)	**0.8339**	0.8338	0.8292

Table 2: Weighted AUC. Parenthesised digits are the seed standard deviation in units of the last decimals. The pre-trained arms lead at every fraction ≥ 0.01; only at 0.003 does the ordering invert.

Three readings:

1. **Pre-training wins, decisively.** At fractions 0.01–1.0 both pre-trained arms exceed scratch by 0.014–0.017 in AUC against seed standard deviations of 0.001–0.008 — many standard deviations, and monotone across the grid.

2. **The gain is worth ≈3–4.5× the data.** Interpolating scratch's curve logarithmically in fraction, finetune's ensemble AUC at 0.01, 0.02, 0.05 and 0.1 requires scratch fractions of 0.033, 0.079, 0.222 and 0.437 respectively, i.e. multipliers 3.3×, 4.0×, 4.4×, 4.4×; at full statistics scratch never reaches finetune's value at all.

3. **Frozen ≈ finetune everywhere** (differences ≤ 0.004, mostly within seed errors). The benefit lives in the pre-trained *representation*; adapting the backbone adds nothing measurable. This is the sharpest single answer of the trial to the "did the pretext retain κ_λ-relevant continuous information?" question: it did.

The 0.003 inversion. At the smallest fraction scratch leads (ensemble 0.7738 vs. 0.7576/0.7699). That tier is in a pathological optimisation regime rather than a representational one: with ≈ 2.5k training events and a global batch of ≈ 8k across 4 GPUs, one "epoch" is a single gradient step, and the selected checkpoints sit at epochs 4–21, i.e. the models are compared after $\mathcal{O}(10)$ updates under *different* learning-rate recipes (the fine-tune recipe uses an aggressive head learning rate over a small body one; the scratch recipe is uniform). We regard this tier as a recipe artefact pending a batch-size/learning-rate ablation, not as evidence about representations.

5 Result 2 — calibration (gate G0.5)

Verdict under (P1)/(P2): a null. The first-passing fraction is `none` for all three arms, so the closure left-shift — the pre-registered adoption metric — is undefined. Under the (P4) fallback, the arms overlap within seed errors and *trade places*: scratch is nominally best at 0.01–0.05 (largest gap at 0.02, ≈2.3σ), the pre-trained arms at 0.1 and 1.0. There is no consistent ordering and no left-shift.

The failure is statistics-limited, not fixed. $\mathrm{SC}_{\mathrm{norm}}$ falls monotonically with training fraction (0.32 → 0.07 finetune, 0.47 → 0.04 frozen, 0.35 → 0.08 scratch) and χ^2/ndf drops from 60–190 to 6–15. The estimator does converge; it simply has not reached the 1%/5% gates by 662k training events in any arm.

	$\lvert\mathrm{IC}\rvert$ (seed mean; stat. error $\approx$0.01–0.02)			$\mathrm{SC}_{\mathrm{norm}}$ (seed mean $\pm$ s.d.)		
fraction	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.130	0.468	0.202	0.318(66)	0.470(79)	0.348(109)
0.01	0.205	0.361	0.237	0.343(123)	0.337(160)	**0.207(172)**
0.02	0.292	0.206	0.205	0.273(81)	0.235(77)	**0.170(56)**
0.05	0.188	0.215	0.124	0.165(41)	0.195(46)	**0.158(58)**
0.1	0.229	0.272	0.332	0.164(33)	**0.154(64)**	0.165(69)
1.0	0.133	0.171	0.156	0.074(56)	**0.044(44)**	0.084(48)

Table 3: Closure, single models. Gates: $\lvert\mathrm{IC}\rvert \le \max(0.01, 3\sigma_{\mathrm{stat}})$ and shape ≤ 0.05. **No arm passes at any fraction.**

Ensembling does not rescue it. Because $\mathbb{E}_{\mathrm{ref}}$ is linear in r, the ensemble ratio $\bar{r} = \frac{1}{K}\sum_k r_k$ satisfies the identity $\mathrm{IC}_{\mathrm{ens}} = \frac{1}{K}\sum_k \mathrm{IC}_k$ exactly: the ensemble carries no information beyond the *signed* mean of the members' offsets. The measurement is unambiguous — in 15 of 18 cells the per-seed mean $\lvert\mathrm{IC}\rvert$ and the signed mean are numerically identical (e.g. finetune 0.02: 0.2917 vs. 0.2917; frozen 1.0: 0.1706 vs. 0.1706), i.e. **all five seeds miss in the same direction**. The offsets are a same-sign systematic ($\mathbb{E}_{\mathrm{ref}}[\hat{r}] - 1 \approx +0.13$ to $+0.29$), not zero-centred model noise, so there is nothing for averaging to cancel. (Ensembling does help in the three mixed-sign cells: finetune 0.003, $0.130 \to 0.038$; scratch 0.01, $0.237 \to 0.047$; scratch 0.05, $0.124 \to 0.043$.) Ensemble AUC improves slightly for all arms and preserves the FM advantage (Table 2).

6 The ceiling comparison

The feature-level BDT was measured on the same physics at full statistics with class-normalised weights (so its prior factor is unity and its closure numbers are directly comparable):

fraction	AUC	$\lvert\mathrm{IC}\rvert$	shape
0.01	0.7917	0.202	0.391
0.1	0.8130	0.011	0.039
1.0	0.8174	0.018	0.000

Table 4: Kinematic ceiling (10 published features, gradient-boosted trees).

Two comparisons matter, and they point in opposite directions:

- **Discrimination: EveNet wins.** At 0.1 and 1.0 both pre-trained arms exceed the ceiling (0.821/0.822 vs. 0.813; 0.834 vs. 0.817), i.e. the object-level model extracts κ_λ information the 10 aggregate features discard. *Caveat:* the ceiling was measured on the un-cut flat-feature population, whereas the sweep applies the valid-b-pair requirement, so the sweep population is slightly easier; the margin at 1.0 (+0.017) is larger than we would expect this effect to explain, but the comparison carries an asterisk that the arm-to-arm comparisons do not.

- **Calibration: the BDT wins by an order of magnitude.** At full statistics the ceiling reaches $\lvert\mathrm{IC}\rvert = 0.018$ and shape 0.000; every EveNet arm sits at $\lvert\mathrm{IC}\rvert = 0.13$–$0.17$. *A simple boosted-tree ensemble with worse AUC produces a far better-calibrated ratio.* This is the observation that shifts the interpretation from "this task is hard to calibrate" to "this training objective does not calibrate" (Sec. 7).

7 Interpretation: a dissociation, and a candidate mechanism

The headline. Pre-training improved *ordering* substantially and *calibration* not at all. These are separable properties: AUC is invariant under any monotone remap of the score, and an entire family of classifiers shares a given AUC while only one member is the density ratio. A method evaluated solely by significance-improvement characteristic (SIC) or background rejection — as in the EveNet paper [1] and the controlled objective study [2] respectively — would report an unambiguous success here and would be structurally unable to see the calibration null. For NSBI, where the per-event ratio enters the likelihood, the calibration axis is the binding one.

Candidate mechanism (testable). EveNet's classification head is trained under diffusion-time conditioning: the loss is $\alpha^2(t)\,\mathrm{CE}(y, \hat{y})$ with a noising probability that our configuration — following the authors' own downstream recipe — sets to 1.0, while prediction is performed at the clean endpoint. A t-averaged cross-entropy is *not* a proper scoring rule for the $t = 0$ posterior: its minimiser preserves the ranking of events (hence the strong AUC) while the logit *scale* need not equal that of the clean-input posterior (hence the systematic $\mathbb{E}_{\mathrm{ref}}[\hat{r}] > 1$). Three observations are consistent with this and inconsistent with a residual convention error: the offsets are same-sign across seeds and arms *including scratch*; they vary cell-to-cell rather than being a common constant (a wrong prior factor would give an identical IC everywhere); and they correlate cell-by-cell with $\mathrm{SC}_{\mathrm{norm}}$ (frozen 0.003: 0.47/0.31; frozen 1.0: 0.17/0.00), as expected if both are driven by per-event log-ratio noise — for which $\mathbb{E}_{\mathrm{ref}}[\hat{r}] \approx e^{\sigma^2/2} > 1$, with $\mathbb{E}_{\mathrm{ref}}[\hat{r}] = 1.2$ corresponding to $\sigma \approx 0.6$.

The decisive ablation. Re-run one column of the grid (say fraction 1.0, all three arms, 5 seeds = 15 runs, hours) with the classification noising disabled (`noise_prob: [0.0, 0.0]`), everything else identical. If |IC| collapses toward the ceiling's 0.018, the calibration null is a fixable property of the training recipe and the FM programme continues on a corrected objective. If it does not, the null is a property of the pre-trained representation's transfer to continuous ratio estimation, and the objective-mismatch verdict of the feasibility note stands.

8 Corrections applied during analysis

For reproducibility, three defects were found and fixed *after* predictions existed but *before* the numbers above were read. All are estimator/plumbing fixes, not changes to the decision rule.

1. **Prior convention (largest effect).** The evaluator initially applied $f_{\mathrm{prior}} = W_0/W_1 = 0.556$ (raw-weight training). Scanning the three candidate conventions on all 15 full-statistics runs selected *class-balanced* ($f = 1$) unanimously; EveNet equalises classes internally. The spurious factor had produced $|\mathrm{IC}| \approx 0.30$–$0.47$ *flat in* N and inflated shape closure by the same constant (a scale error shifts every bin equally). This is why the corrected numbers, unlike the first pass, decrease with statistics.

2. **Weight treatment.** Closure now uses $|w|$: binary cross-entropy training cannot consume negative weights, so the reference density the model saw is the $|w|$ one (the reference sample carries 6.1% negative weights).

3. **Predict-path configuration.** Predict configs inherited a relative `options.yaml` path that resolved into the (empty) config farm, and lacked the head-inclusion block, so every prediction task died before loading a checkpoint. Both are now generated from the matching training config.

9 Caveats and scope

- **No sensitivity claim.** No σ_{κ_λ} or absolute-sensitivity statement is made or implied; the stock schema discards τ-identification and FastMTT $m_{\tau\tau}$ (equally in all arms), and the MC-statistics estimator question remains open upstream [3].
- **One channel, one hypothesis pair.** κ_λ= 1 vs. κ_λ= 5, $\mu\tau_h$ and $e\tau_h$ only. The κ_λ= 0 pair and the systematic-variation task (Money Plot 2, nominal vs. injected JES(m_{HH})) are untouched.
- **Population mismatch vs. the ceiling only** (Sec. 6); all arm-to-arm comparisons share identical events.
- **Five seeds** give seed errors of 0.03–0.17 on $\mathrm{SC}_{\mathrm{norm}}$, which is why the closure comparison is a null rather than a measured equality.
- **The 0.003 tier** is optimiser-transient-dominated (Sec. 4).
- One frozen cell (0.02, seed 3) and one scratch run (0.01, seed 2 — early-stopped at epoch 1) are degraded; both inflate their cells' seed spread and neither changes a conclusion.

10 Implications and next steps

For the FM programme. G0 passes on discrimination with a 3–4.5× data multiplier; G0.5 is a null. Under the pre-registered rule (P1) that combination is *not* adoption for NSBI — but it is also not the objective-mismatch verdict the feasibility note feared, because the representation demonstrably carries κ_λ-relevant information (frozen ≈ finetune, both above the ceiling on AUC). The programme's decision therefore hinges on the ablation of Sec. 7, which is cheap.

Priorities.

N1. **[P0, S]** Noising ablation at fraction 1.0 (15 runs) — decides whether the calibration null is recipe or representation.

N2. **[P1, S]** G1 corner-encoding A/B (τ_h at the vacant flag corner): the NPZs already exist for both encodings.

N3. **[P1, M]** Money Plot 2 (systematic-variation data efficiency): the FM's low-statistics claim aimed at nuisance parameterisation, using the analytic injector where closure is exact by construction.

N4. **[P2, S]** Low-fraction recipe ablation (batch size / learning rate at 0.003) to convert the inversion from artefact to measurement.

N5. **[P2, S]** Re-measure the ceiling on the valid-b-pair population to remove the asterisk in Sec. 6.

For the EveNet authors (discussion, 16 July 2026). Three concrete items: (i) is a calibration-preserving classification mode available or planned — i.e. a $t = 0$-only classification loss — given that the ranking transfers but the calibrated ratio does not? (ii) our G1 result will be the first data on reusing *vacant flag corners* for object types absent from pre-training (τ_h, γ); would they bless this as the zero-surgery interim, and (iii) would they consider retrofitting an explicit type-embedding table — initialised from the current flag patterns so it is function-preserving on day one — so that future object types become an append-a-row operation rather than schema surgery?

References

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